

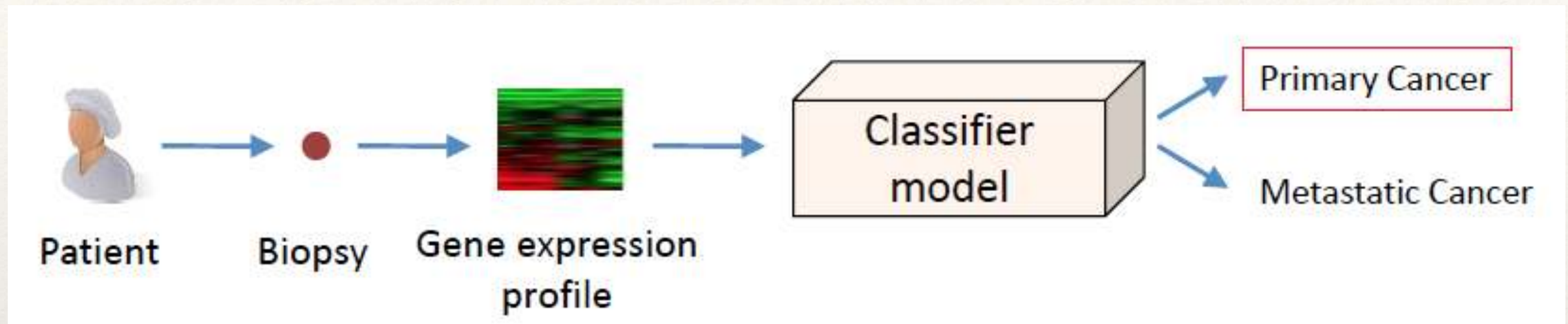
Properties of SVM

- Flexibility in choosing a similarity function
- **Sparseness** of solution when dealing with large data sets
 - only support vectors are used to specify the separating hyperplane
- Ability to **handle large feature spaces**
 - complexity does not depend on the dimensionality of the feature space
- Overfitting can be controlled by soft margin approach
- **Nice math property**: a simple convex optimization problem which is guaranteed to converge to a single global solution
- Feature Selection

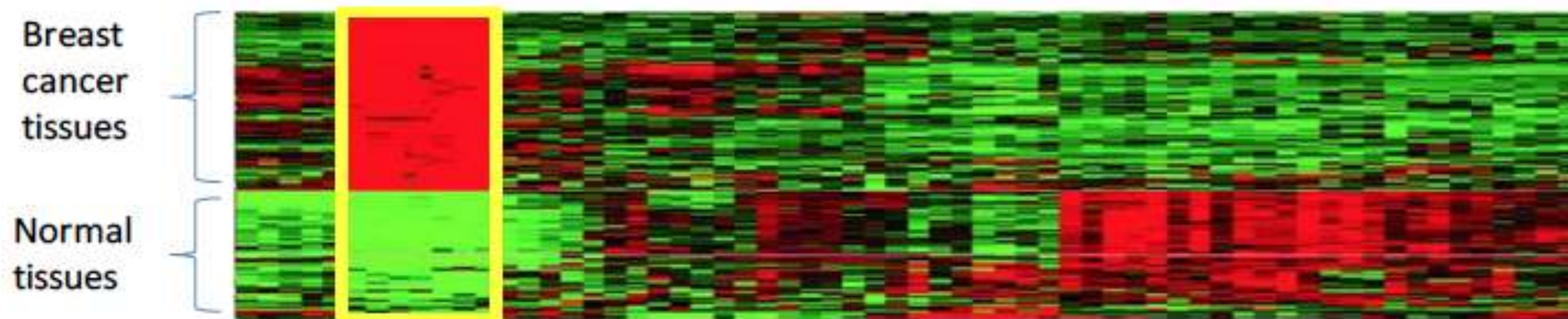
SVM Applications

- SVM has been used successfully in many real-world problems
 - text (and hypertext) categorization
 - image classification
 - bioinformatics (Protein classification, Cancer classification)
 - hand-written character recognition

Application 1: Cancer Classification

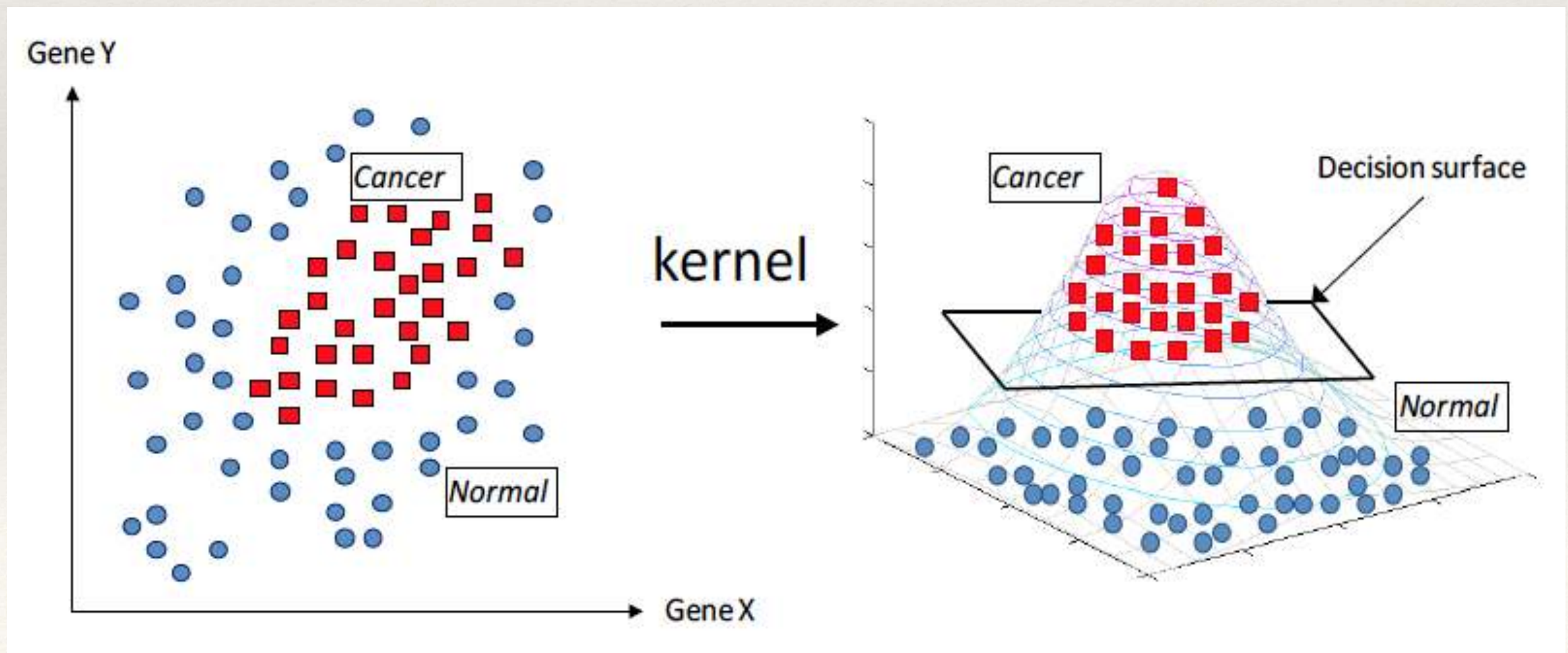
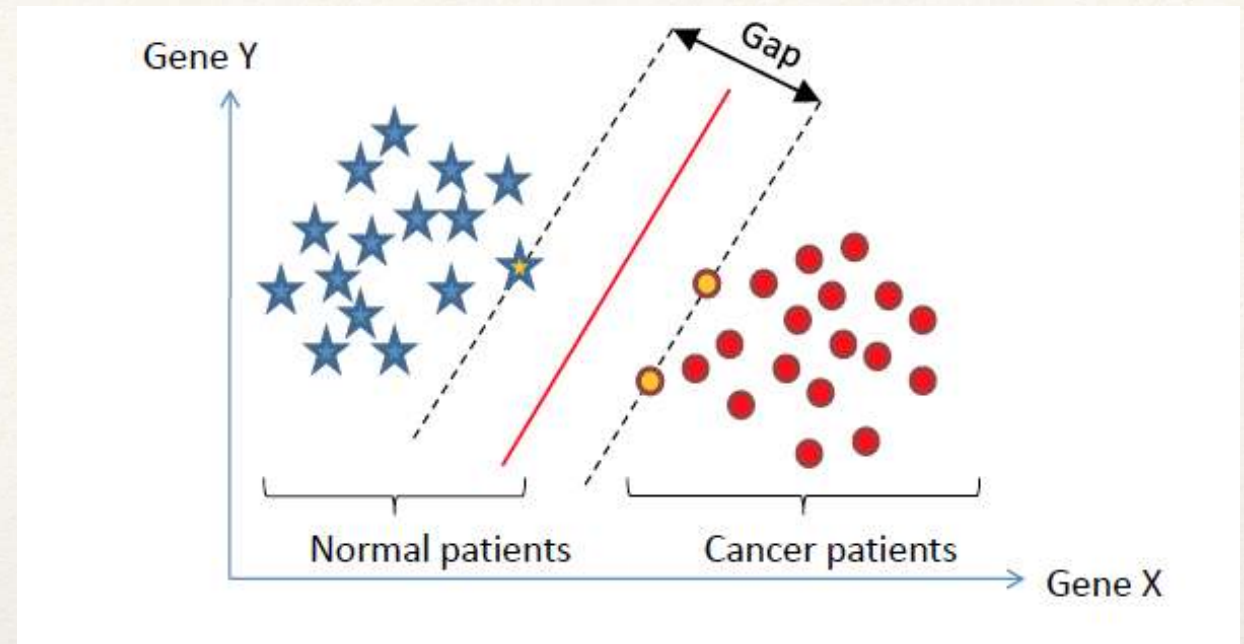


- E.g., find the most compact panel of breast cancer biomarkers from microarray gene expression data for 20,000 genes:

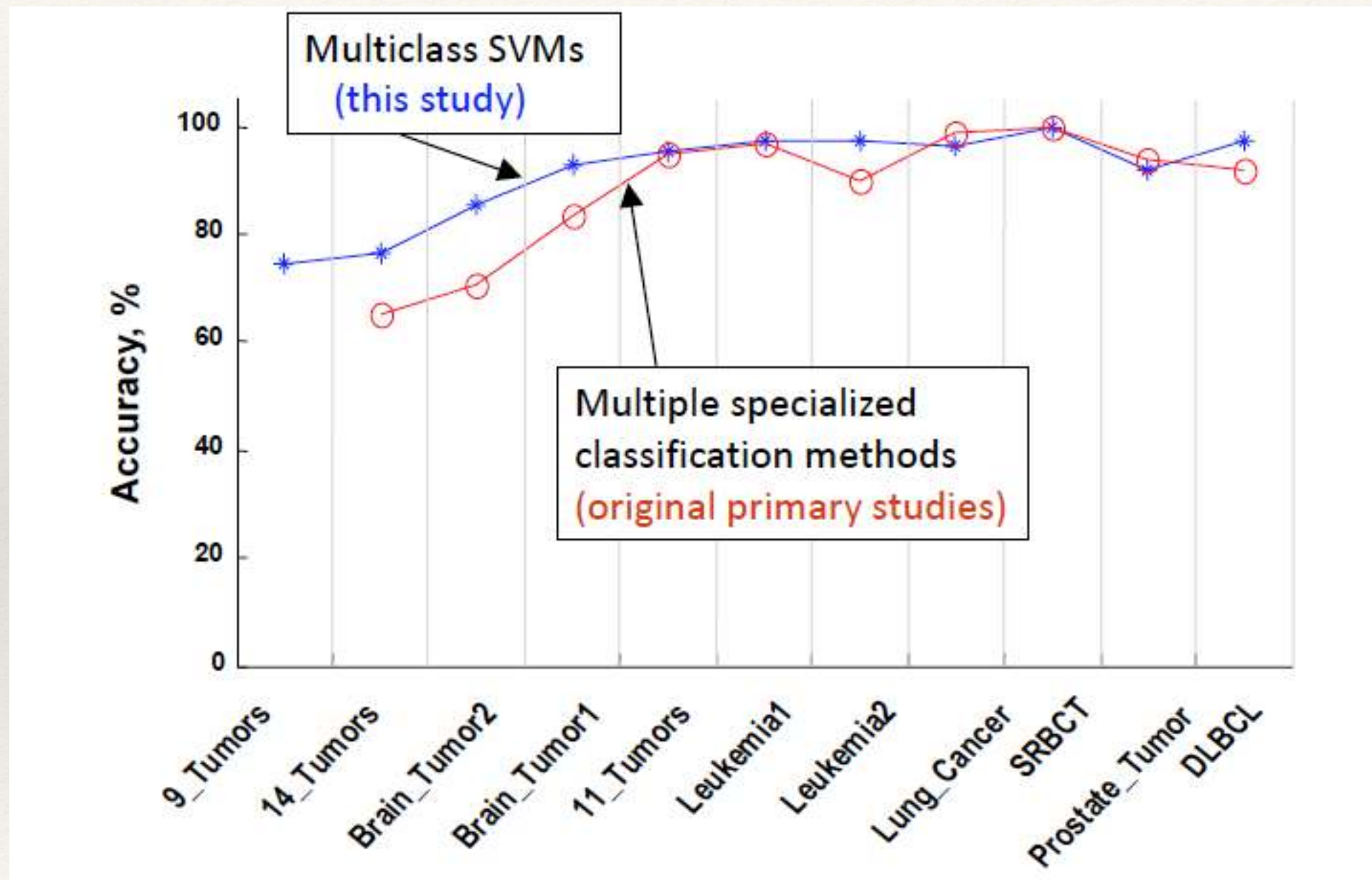


Application 1: Cancer Classification

Linear Versus Non-linear SVMs



Application 1: Cancer Classification



Weakness of SVM

- **It is sensitive to noise**
 - A relatively small number of mislabeled examples can dramatically decrease the performance
- **It only considers two classes**
 - how to do multi-class classification with SVM?
 - Answer:
 - 1) with output m , learn m SVM's
 - SVM 1 learns "Output==1" vs "Output != 1"
 - SVM 2 learns "Output==2" vs "Output != 2"
 - :
 - SVM m learns "Output== m " vs "Output != m "
 - 2) To predict the output for a new input, just predict with each SVM and find out which one puts the prediction the furthest into the positive region.

Application 2: Text Categorization

- Task: The classification of natural text (or hypertext) documents into a fixed number of predefined categories based on their content.
 - email filtering, web searching, sorting documents by topic, etc..
- A document can be assigned to more than one category, so this can be viewed as a series of binary classification problems, one for each category.

Application 2: Text Categorization

IR's vector space model (aka bag-of-words representation)

- A doc is represented by a vector indexed by a pre-fixed set or dictionary of terms
- Values of an entry can be binary or weights

$$\phi_i(x) = \frac{\text{tf}_i \log(\text{idf}_i)}{\kappa},$$

- Doc $x \Rightarrow \phi(x)$

Application 2: Text Categorization

- The distance between two documents is $\langle \phi(x) \phi(z) \rangle$
- $K(x,z) = \langle \phi(x) \phi(z) \rangle$ is a valid kernel, SVM can be used with $K(x,z)$ for discrimination.
- Why SVM?
 - High dimensional input space
 - Few irrelevant features (dense concept)
 - Sparse document vectors (sparse instances)
 - Text categorization problems are linearly separable

Application 2: Text Categorization

					SVM (poly) degree $d =$					SVM (rbf) width $\gamma =$			
	Bayes	Rocchio	C4.5	k-NN	1	2	3	4	5	0.6	0.8	1.0	1.2
earn	95.9	96.1	96.1	97.3	98.2	98.4	98.5	98.4	98.3	98.5	98.5	98.4	98.3
acq	91.5	92.1	85.3	92.0	92.6	94.6	95.2	95.2	95.3	95.0	95.3	95.3	95.4
money-fx	62.9	67.6	69.4	78.2	66.9	72.5	75.4	74.9	76.2	74.0	75.4	76.3	75.9
grain	72.5	79.5	89.1	82.2	91.3	93.1	92.4	91.3	89.9	93.1	91.9	91.9	90.6
crude	81.0	81.5	75.5	85.7	86.0	87.3	88.6	88.9	87.8	88.9	89.0	88.9	88.2
trade	50.0	77.4	59.2	77.4	69.2	75.5	76.6	77.3	77.1	76.9	78.0	77.8	76.8
interest	58.0	72.5	49.1	74.0	69.8	63.3	67.9	73.1	76.2	74.4	75.0	76.2	76.1
ship	78.7	83.1	80.9	79.2	82.0	85.4	86.0	86.5	86.0	85.4	86.5	87.6	87.1
wheat	60.6	79.4	85.5	76.6	83.1	84.5	85.2	85.9	83.8	85.2	85.9	85.9	85.9
corn	47.3	62.2	87.7	77.9	86.0	86.5	85.3	85.7	83.9	85.1	85.7	85.7	84.5
microavg.	72.0	79.9	79.4	82.3	84.2	85.1	85.9	86.2	85.9	86.4	86.5	86.3	86.2
					combined: 86.0					combined: 86.4			

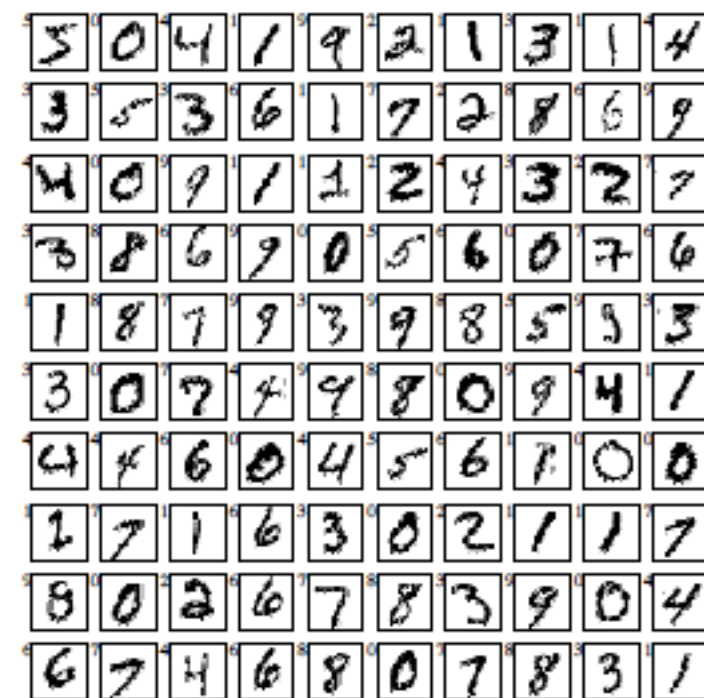
Application 3: Handwriting Recognition

For example MNIST hand-writing recognition.

60,000 training examples, 10000 test examples, 28x28.

Linear SVM has around 8.5% test error.

Polynomial SVM has around 1% test error.



SVMs : full MNIST results

Classifier	Test Error
linear	8.4%
3-nearest-neighbor	2.4%
RBF-SVM	1.4 %

Some Considerations

- Choice of kernel
 - Gaussian or polynomial kernel is default
 - if ineffective, more elaborate kernels are needed
 - domain experts can give assistance in formulating appropriate similarity measures
- Choice of kernel parameters
 - e.g. σ in Gaussian kernel
 - σ is the distance between closest points with different classifications
 - In the absence of reliable criteria, applications rely on the use of a validation set or cross-validation to set such parameters.
- Optimization criterion – Hard margin v.s. Soft margin
 - a lengthy series of experiments in which various parameters are tested

Software

30 SVMs : software

Lots of SVM software:

- LibSVM (C++)
- SVMLight (C)

As well as complete machine learning toolboxes that include SVMs:

- Torch (C++)
- Spider (Matlab)
- Weka (Java)

All available through www.kernel-machines.org.